

Project Summary Report: Lead Conversion Model

1. Project Overview and Problem Statement

The primary objective of this assignment was to enhance the sales efficiency of X Education by developing a predictive model to identify 'Hot Leads'—those most likely to convert into paying customers. The company faces a low overall conversion rate (approx 37%) and aims to optimize its sales calls by focusing on high-potential leads. The core requirement was to build a Logistic Regression model to assign a Lead Score (0-100) to each prospect.

2. Analytical Approach and Data Handling

The analysis began with cleaning the provided dataset of approx 9000 leads. Crucial steps included treating the 'Select' entries in categorical columns as null values and subsequently dropping columns with a high volume of missing data (e.g., Lead Quality, Asymmetrique indices). Remaining missing values were imputed using mode/median, and outliers in numerical features like TotalVisits were capped.

Feature engineering involved converting all relevant categorical variables into dummy variables and scaling numerical features to ensure all inputs contributed fairly to the model training. The data was split into a 70-30 train-test set.

3. Model Development and Evaluation

A Logistic Regression model was developed using the statsmodels library. Feature selection was managed using Recursive Feature Elimination (RFE) combined with manual elimination based on high p-values and checking for multicollinearity using the Variance Inflation Factor (VIF). The final model contained 14 statistically significant variables.

The model achieved excellent performance metrics on the test set: an **ROC AUC Score of 0.90** and an overall **Accuracy of approx 85%**.

4. Optimal Cut-off and Lead Score

To translate the predicted probability into a binary decision, an optimal cut-off probability was determined by plotting the trade-off between Sensitivity (true positive rate) and Specificity (true negative rate). The balanced cut-off was found to be **0.35**.

- **Lead Score:** Assigned as Probability x 100.
- **Initial Classification:** Leads with a score > 35 are designated as 'Hot Leads'.

At this cut-off, the model achieved a **Sensitivity of approx 85%** and a **Precision (conversion rate among 'Hot Leads') of approx 73.6%**. This conversion rate is very close to the CEO's ballpark target of 80%, confirming the model's ability to isolate high-value leads.

5. Business Insights and Lessons Learned

The model revealed key drivers for conversion:

1. **Strongest Positive Predictor:** Leads identified as **Working Professionals**.
2. **Strongest Negative Predictor:** Leads who selected **'Do Not Email'**.

The primary lesson learned is the importance of adapting the probability cut-off to meet varying business objectives. A lower cut-off is ideal for maximizing total conversions (e.g., during the intern period), as it reduces the costly Type II error (missing a sale). Conversely, a higher cut-off is necessary to maximize sales team efficiency (minimizing useless calls/False Positives) when quarterly targets are already met.

In conclusion, the Logistic Regression model provides X Education with a robust, data-driven tool to prioritize sales efforts and improve its lead conversion rate significantly.